

Chronic Disease Surveillance

Test Data Package — Folder Contents Documentation

1. Package Structure

The complete test data is delivered as a single zip file:

chronic_disease_test_data.zip

The zip contains three subfolders, one per CMS quality measure:

Subfolder	CMS Measure	Folders
poor-diabetic-control /	CMS122 — Diabetes HbA1c Poor Control ($\geq 9\%$)	365
controllabe-bp/	CMS165 — Controlling High Blood Pressure	528

2. Folder Contents

Each scenario folder contains files for the input data, the expected evaluation result, and the complete reporting package.

2.1 Standard Folder (4 files)

When the patient qualifies for the Initial Population, each folder contains:

File Pattern	Format	Purpose
{scenario}.json	FHIR Bundle (collection)	Input test data: Patient, Encounter, Condition, Observation, and any MedicationRequest, ServiceRequest, or Procedure resources
MeasureReport_{uuid}.json	FHIR MeasureReport	Expected evaluation result with population counts (IP, Denominator, Numerator, Exclusion, Exception) and quality Score
Bunlde_{uuid}.json	FHIR Bundle (message)	Electronic Case Report (eCR): MessageHeader + nested Bundle combining all input data with the MeasureReport

2.2 Not-In-Population Folder (2 files)

When the patient does not meet Initial Population criteria, no eCR is generated:

File Pattern	Format	Purpose
{scenario}.json	FHIR Bundle (collection)	Input test data (same structure as standard folder)
MeasureReport_{uuid}.json	FHIR MeasureReport	All population counts are zero (IP=0, Den=0, Num=0)

3. Input Bundle Resources

The input {scenario}.json is a FHIR R4 Bundle (type: collection) containing 3–7 resources:

Resource	Present	Description
Patient	Always	Demographics, MRN identifier, US Core race/ethnicity extensions
Encounter	Always	Office visit (CPT 99213/99214/99215/99404), status finished, period within 2025 MP
Condition	Always	Primary diagnosis per measure. May include comorbidities (HTN, CKD, Obesity, CAD, Bipolar, Pregnancy)
Observation	Most	HbA1c lab result (CMS122), BP panel with systolic/diastolic (CMS165), or depression screening (CMS2)
MedicationRequest	Some	Antidepressant follow-up (CMS2) or diabetes medication. RxNorm coding, minimal fields
ServiceRequest	Some	Mental health referral (CMS2). SNOMED coding
Procedure	Some	Mental health assessment (CMS2) or dialysis (CMS165). SNOMED or CPT coding

4. eCR Reporting Bundle Structure

The create-report files package the input data and evaluation result into a single submission unit:

Entry	Resource	Contents
1	MessageHeader	Routing metadata: event coding, source endpoint, focus reference to the content bundle
2	Bundle (nested)	Collection bundle containing all original input resources (Patient, Encounter, Condition, Observation, etc.) plus the MeasureReport

Provided in both JSON and XML formats with identical content.

5. MeasureReport Population Fields

Each MeasureReport contains these population counts:

Field	Value	Meaning
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initial-population	0 or 1	Patient meets age, encounter, and diagnosis criteria for the measure
denominator	0 or 1	Patient is in the Initial Population (equals IP for these measures)
numerator	0 or 1	Measure condition is met (HbA1c $\geq$ 9%, BP controlled, screening completed)
denominator-exclusion	0 or 1	Patient has an exclusion condition (hospice, pregnancy, ESRD, frailty, etc.)
denominator-exception	0 or 1	Screening not performed with documented reason (CMS2 only: declined, medical reason)

6. Example: Folder Contents

Standard Folder

CMS122_DENOM_HbA1c_7p5_GoodControl/

File	Size	Contents
CMS122_DENOM_HbA1c_7p5_GoodControl.json	5.8 KB	Patient + Condition (DM) + Encounter + Observation (HbA1c 7.5%)
MeasureReport_c860c166-...60.json	25.5 KB	IP=1, Den=1, Num=0, DenExcl=0, Score=1.0
Bundle_016788c3-..._718b5dbe-...json	48.0 KB	eCR Bundle: MessageHeader + nested Bundle with all data + MeasureRepor

Not-In-Population Folder

CMS122_NOT_IP_AgeUnder18/

File	Size	Contents
CMS122_NOT_IP_AgeUnder18.json	5.6 KB	Patient (age <18) + Condition (DM) + Encounter + Observation (HbA1c 9.2%)
MeasureReport_ca84c95c-...0a.json	25.2 KB	IP=0, Den=0, Num=0, DenExcl=0 (outside measure)